Analysis and visualization of large-scale model output with an interactive web-based dashboard

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2020 AWRA Virtual Geospatial Water Technology Conference

BACKGROUND

- CEAP II with SWAT+ models
- Connectivity Dataset
  - Hydrologic Connectivity Dataset (White et al., 2017)
    - 86,000 HUC12s provided by the Watershed Boundary Dataset(WBD) (https://www.usgs.gov/core-science-systems/ngp/national-hydrography/watershed-boundary-dataset?qt-science_support_page_related_con=4#qt-science_support_page_related_con)
    - 3 million National Hydrography Dataset (NHDPlus) stream segments (https://www.usgs.gov/core-science-systems/ngp/national-hydrography/nhdplus-high-resolution)

water MDPI

Technical Note

Development of a Hydrologic Connectivity Dataset for SWAT Assessments in the US

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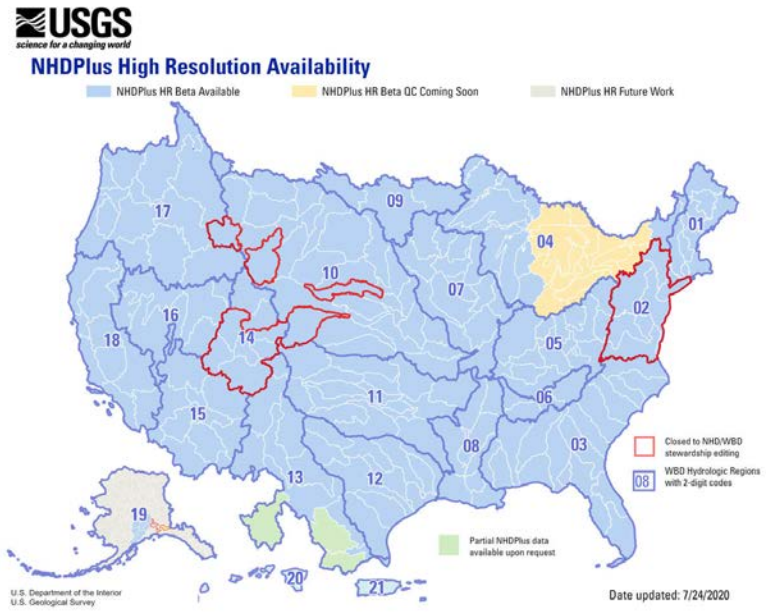
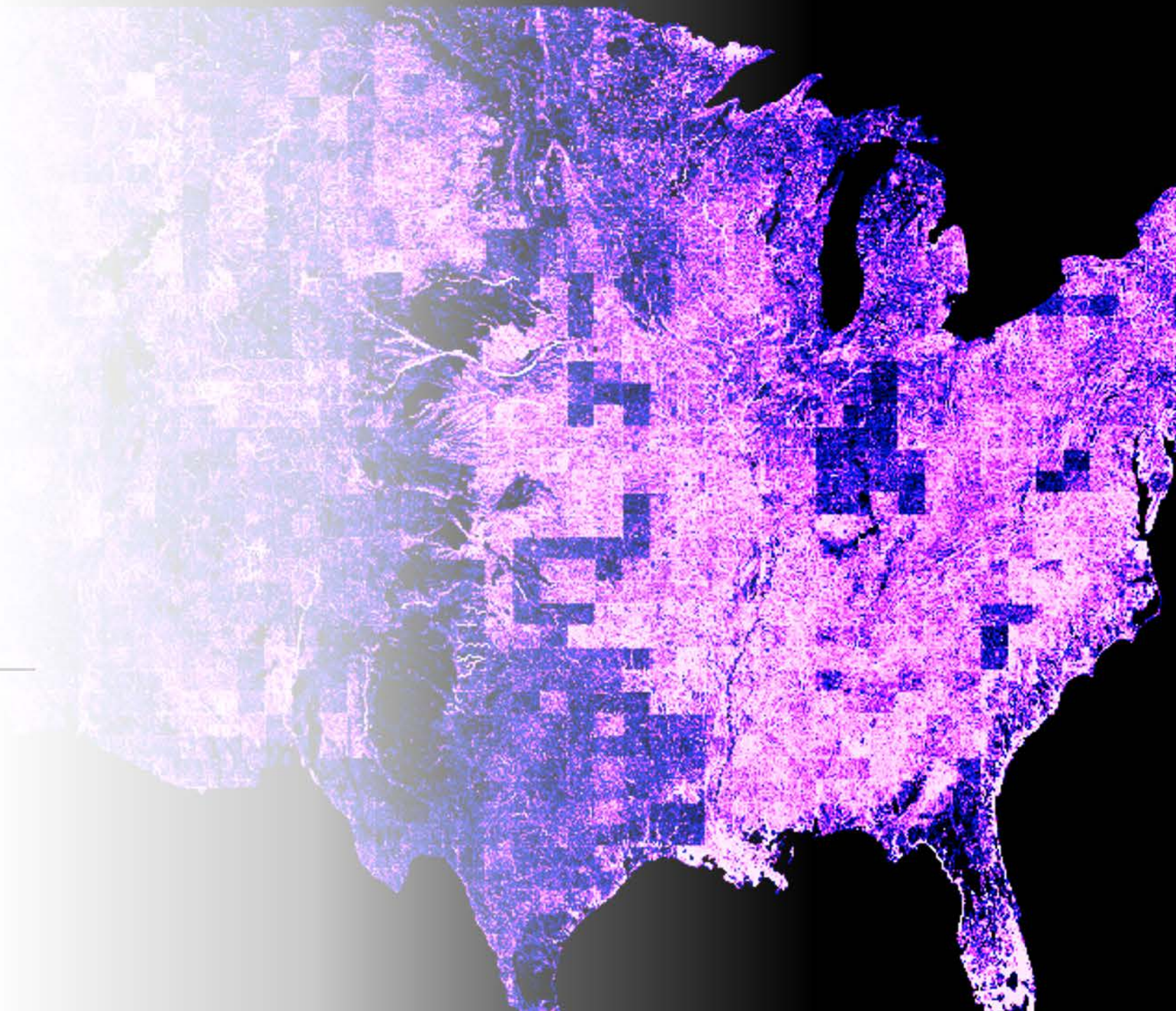
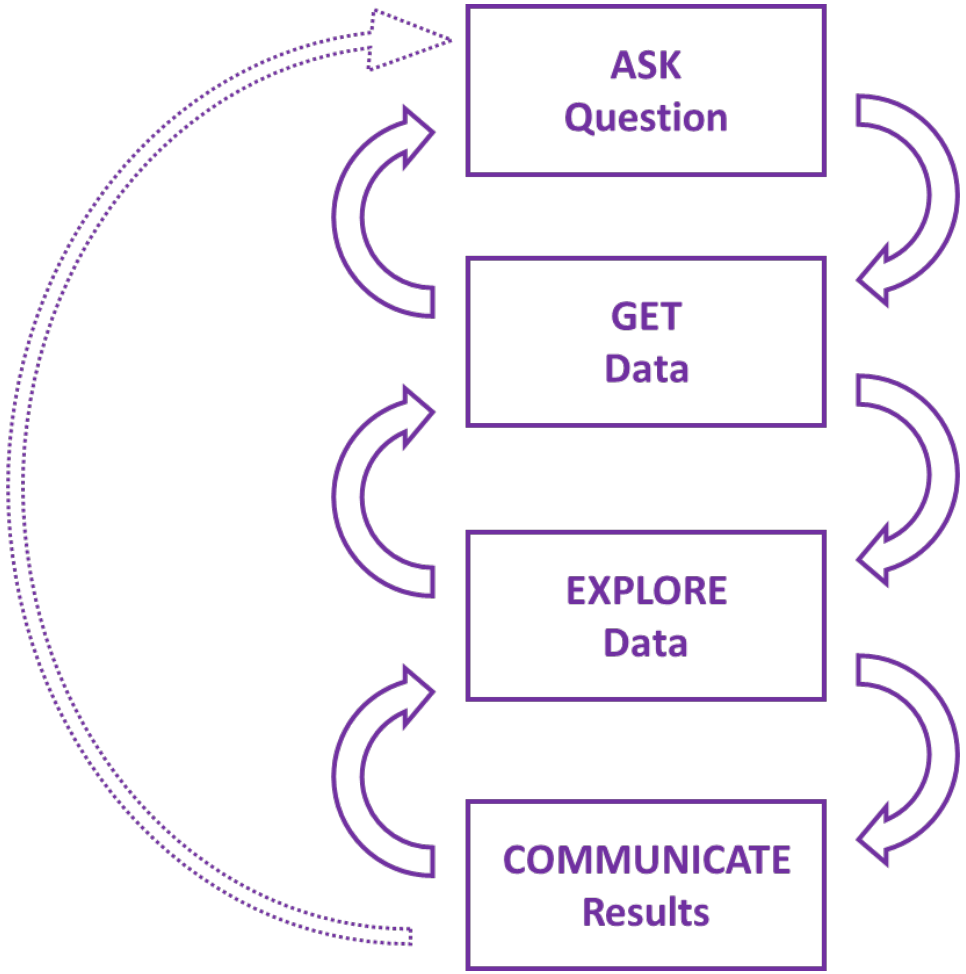
Abstract: Model-based water quality assessments are an important informer of conservation



CHALLENGES



CHALLENGES | WORKFLOW FROM DATA TO DECISION



Source: Dharhas Pothina - Rapid Prototyping and Deployment Using the PyViz Stack and Anaconda Enterprise

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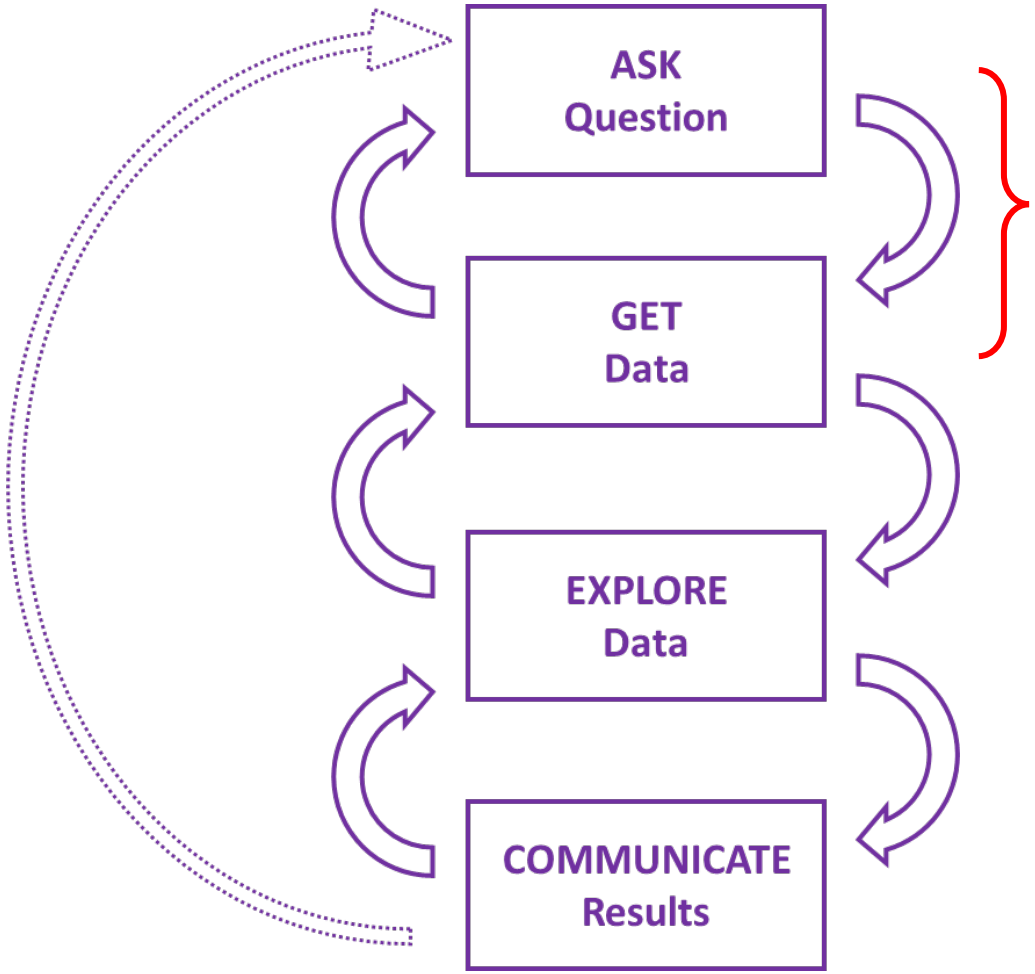
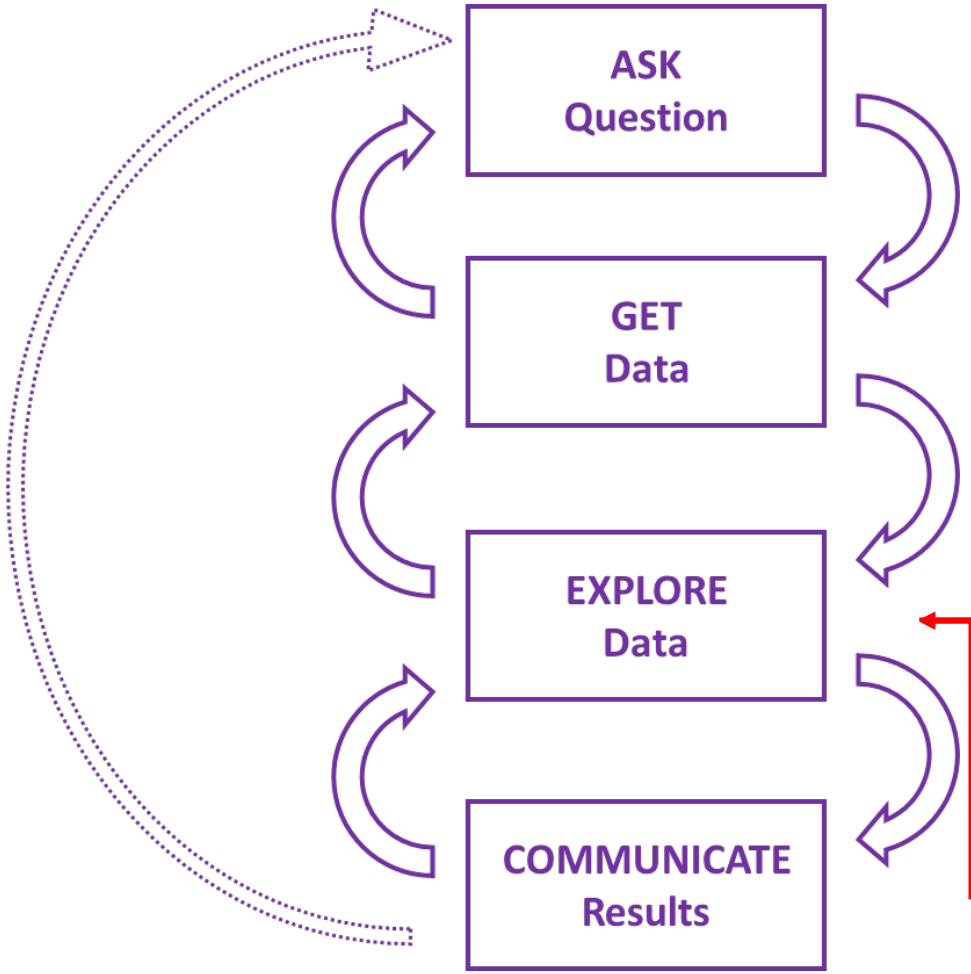


2120 SWAT+ Models

Daily Stream Discharges
(Range in Size between 3 and 4 Terabytes)
Based on a 10-year SWAT+ simulation

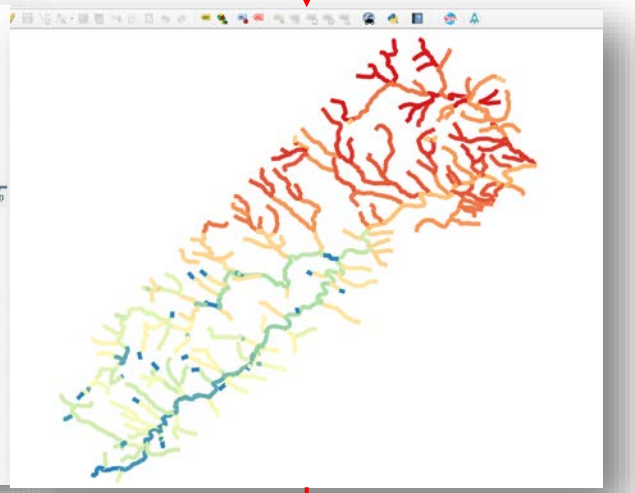
Source: Dharhas Pothina - Rapid Prototyping and Deployment Using the PyViz Stack and Anaconda Enterprise

CHALLENGES | WORKFLOW FROM DATA TO DECISION



Check Connectivity Issue

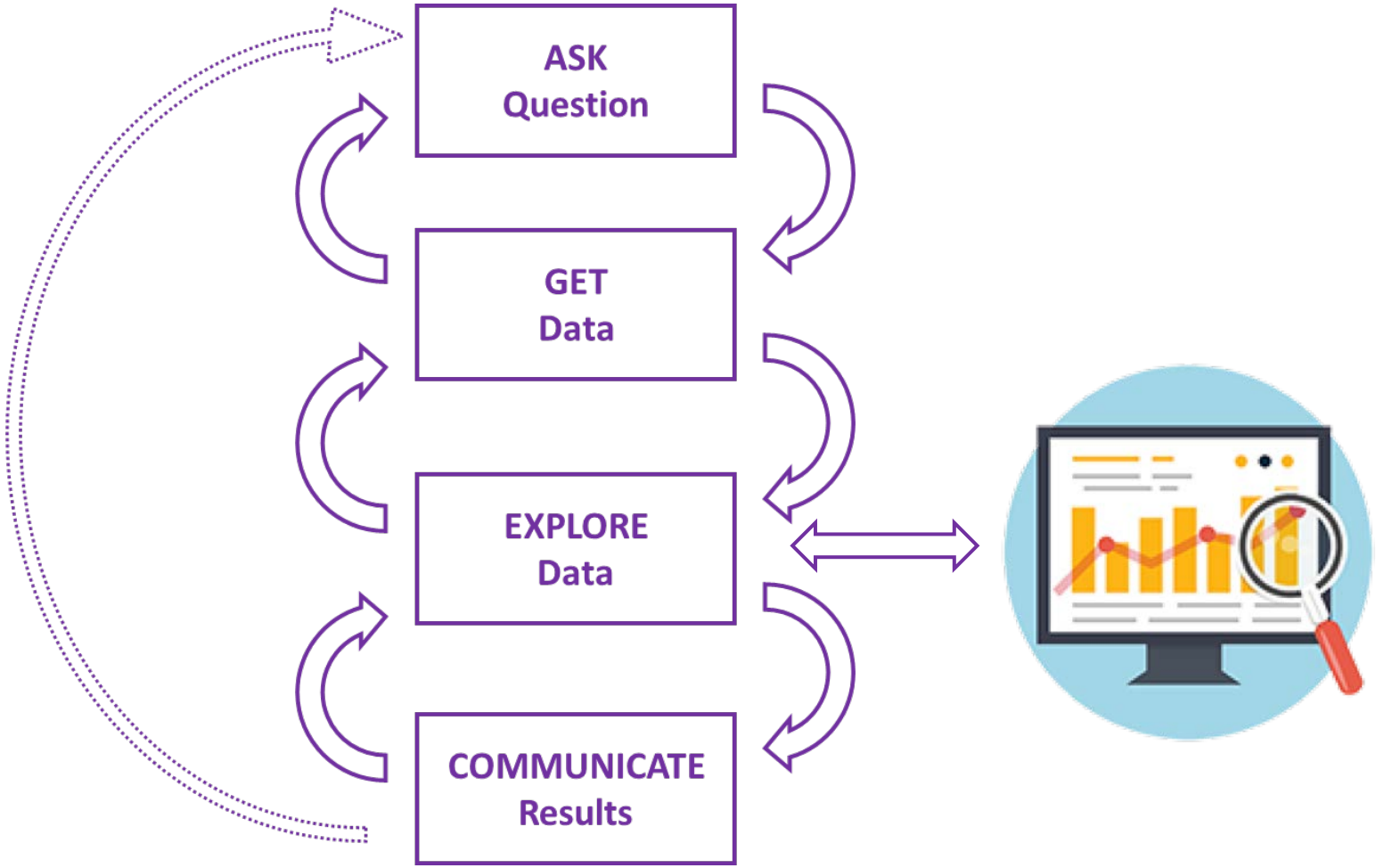
Text Editor / Spread Sheet



- **Data Deluge**
- **Communication Problem**

Source: Dharhas Pothina - Rapid Prototyping and Deployment Using the PyViz Stack and Anaconda Enterprise

CHALLENGES | WORKFLOW FROM DATA TO DECISION



- **Easier to use to communicate results**
- **Easier to explore data**
- **Easier to understand system**

Source: Dharhas Pothina - Rapid Prototyping and Deployment Using the PyViz Stack and Anaconda Enterprise

GOAL

- Show capabilities of HoloViz (PyViz), DASK in Python Ecosystem
- Glimpse into analysis workflow of Scientist

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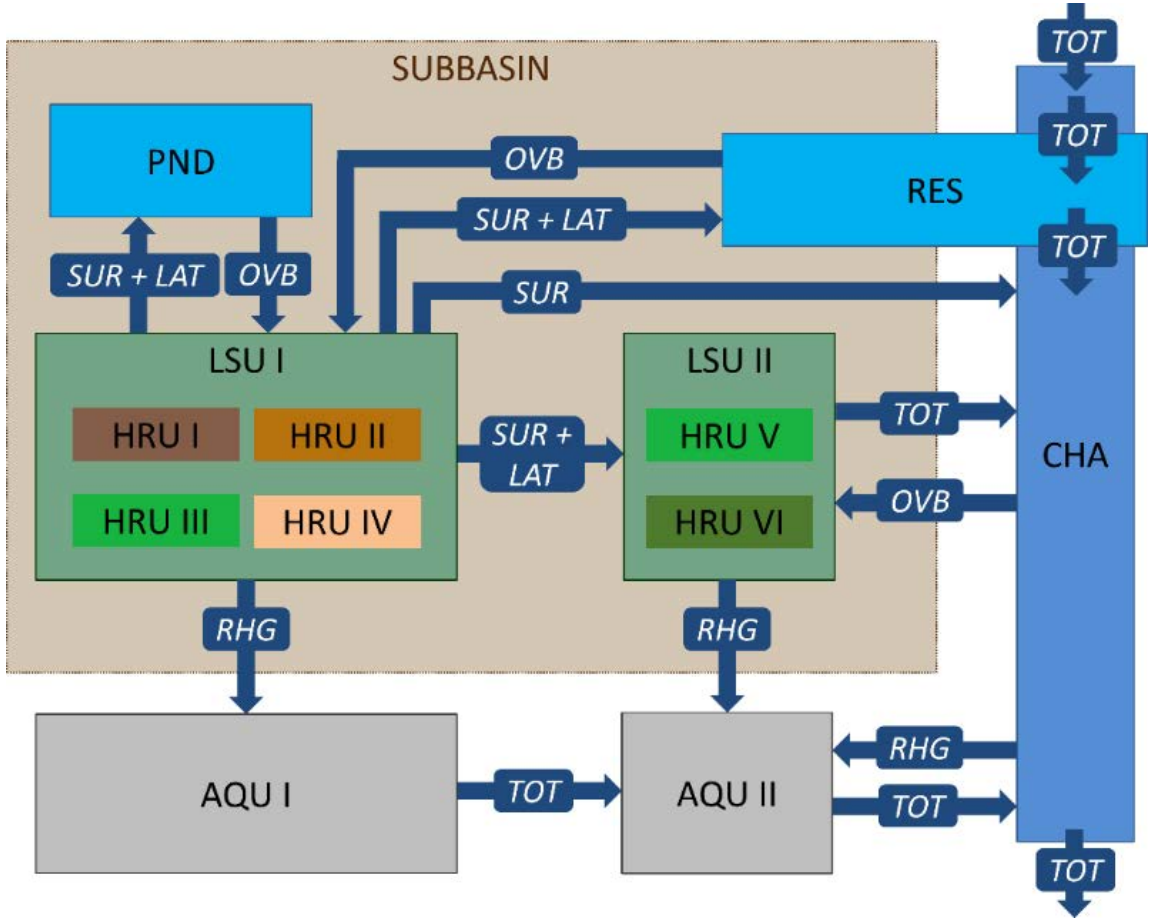
OUTLINE

- What's SWAT+ model? How does simulation outputs look?
- How to use outputs with Python
- How to use them on an interactive dashboard? (Short Demo)
- Q&A

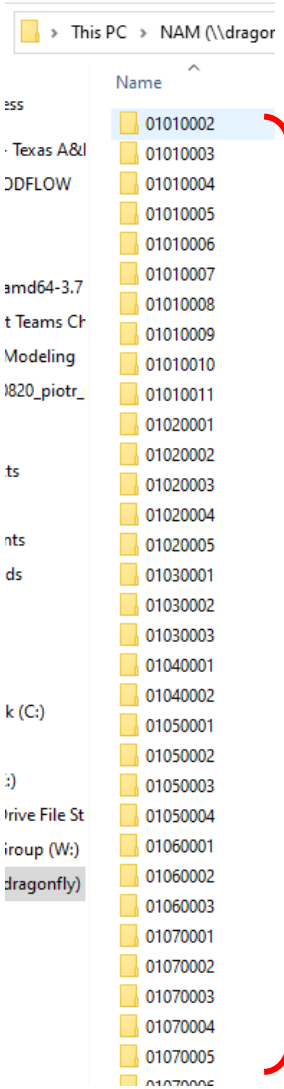
SWAT+ MODEL

- Completely restructured version of SWAT (Bieger et al., 2017)
- SWAT+ Code is restructured as Spatial Object – approach
  - New spatial objects can be defined for external models
  - SWAT+ can easily send output to and receive input from the other models.



Bieger et al.: Introduction to SWAT+, a completely restructured version of the Soil and Water Assessment Tool. Journal of the American Water Resources Association (JAWRA), 2017.

SWAT+ MODEL | SIMULATION OUTPUTS



2120
SWAT+ Models

aquifer_aa.txt
aquifer_yr.txt
basin_crop_yld_aa.txt
basin_crop_yld_yr.txt
basin_ls_aa.txt
basin_ls_yr.txt
basin_nb_aa.txt
basin_nb_yr.txt
basin_pw_aa.txt
basin_pw_yr.txt
basin_sd_cha_aa.txt
basin_sd_cha_yr.txt
basin_sd_chamorph_aa.txt
basin_sd_chamorph_yr.txt
basin_wb_aa.txt
basin_wb_yr.txt
channel_sd_aa.txt
channel_sd_mon.txt
channel_sd_yr.txt
channel_sdmorph_aa.txt
channel_sdmorph_mon.txt
channel_sdmorph_yr.txt
crop_yld_aa.txt
hru_ls_aa.txt
hru_ls_yr.txt
hru_nb_aa.txt
hru_nb_yr.txt
hru_pw_aa.txt
hru_pw_yr.txt
hru_wb_aa.txt
hru_wb_yr.txt
lsunit_ls_aa.txt
lsunit_ls_yr.txt
lsunit_nb_aa.txt
lsunit_nb_yr.txt
lsunit_pw_aa.txt
lsunit_pw_yr.txt
lsunit_wb_aa.txt
lsunit_wb_yr.txt
reservoir_aa.txt
reservoir_yr.txt
wetland_aa.txt
wetland_yr.txt

DAILY, MONTHLY, YEARLY, AVANN / Spatial Scales
≈ 150

viz_conn_lst.py | channel_sdmorph_mon.txt

ceap_01010003 SWAT+ APR 15 2020 MODULAR Rev 2020.60.4

jday	mon	day	yr	unit	gis_id	name	flo_in m^3/s	aqu_in m^3/s	flo_out m^3/s
31	1	31	2000	1	1	t_1_1	0.4233E-06	0.0000E+00	0.1785E-03
31	1	31	2000	2	1	t_1_2	0.1084E-05	0.0000E+00	0.3933E-03
31	1	31	2000	3	1	t_1_3	0.4083E-06	0.0000E+00	0.1921E-03
31	1	31	2000	4	1	t_1_4	0.4650E-06	0.0000E+00	0.1996E-03
31	1	31	2000	5	1	t_2_5	0.1435E-02	0.0000E+00	0.1509E-02
31	1	31	2000	6	1	t_2_6	0.5726E-03	0.0000E+00	0.1243E-02
31	1	31	2000	7	1	t_1_7	0.4539E-06	0.0000E+00	0.1693E-03
31	1	31	2000	8	1	t_2_8	0.1709E-02	0.0000E+00	0.1867E-02
31	1	31	2000	9	1	m_1_9	0.2625E-06	0.0000E+00	0.1002E-03
31	1	31	2000	10	1	t_2_10	0.2037E-02	0.0000E+00	0.2631E-02
31	1	31	2000	11	1	t_1_11	0.1328E-05	0.0000E+00	0.5958E-03
31	1	31	2000	12	1	t_1_12	0.7941E-06	0.0000E+00	0.3239E-03
31	1	31	2000	13	1	t_1_13	0.4612E-06	0.0000E+00	0.1943E-03
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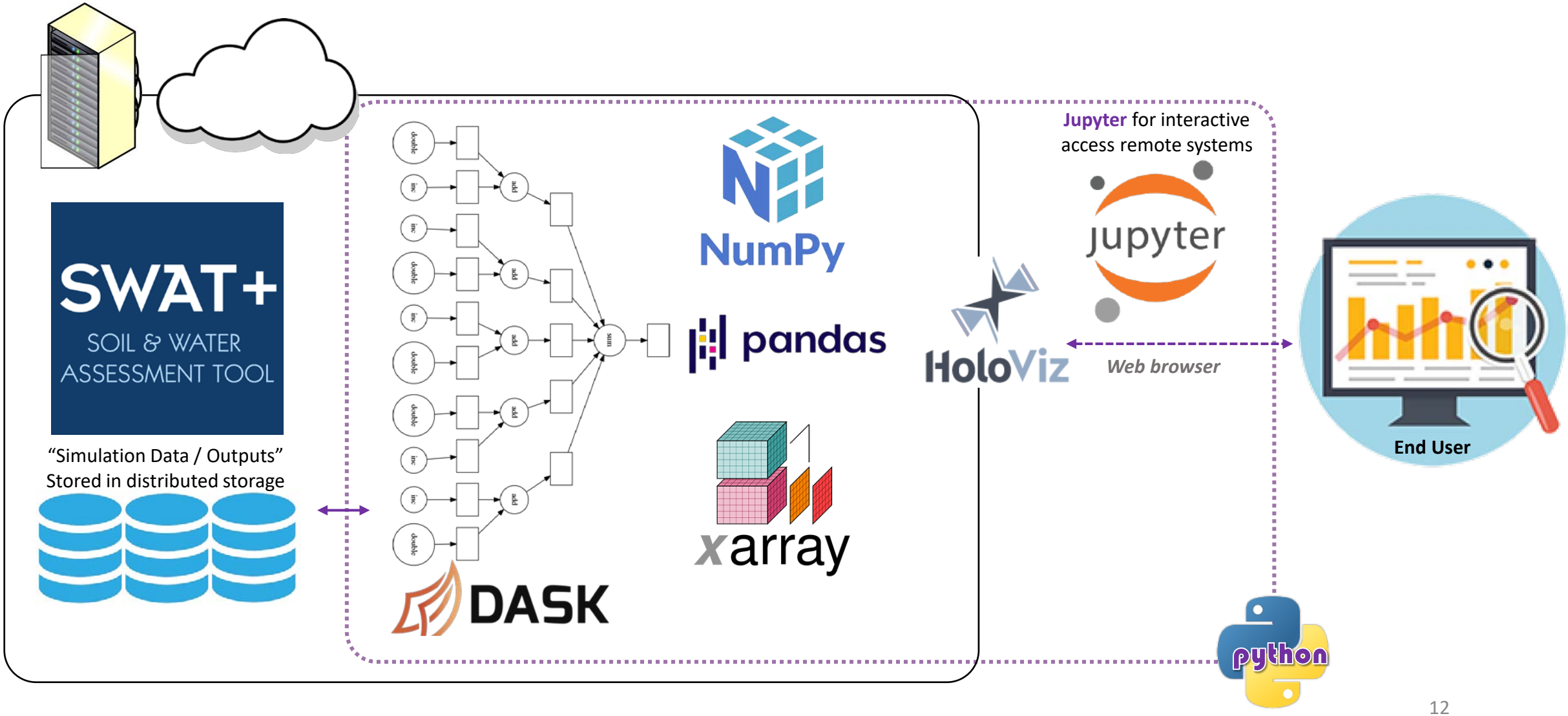
31	1	31	2000	15	1	t_1_15	0.3161E-06	0.0000E+00	0.1427E-03
31	1	31	2000	16	1	t_2_16	0.1612E-02	0.0000E+00	0.1990E-02
31	1	31	2000	17	1	t_1_17	0.5013E-06	0.0000E+00	0.1905E-03
31	1	31	2000	18	1	t_2_18	0.9204E-03	0.0000E+00	0.1421E-02

WORKFLOW


SWAT+
SOIL & WATER ASSESSMENT TOOL
"Simulation Data / Outputs"
Stored in distributed storage
double
inc
add
sum
DASK
NumPy
pandas
xarray
HoloViz
Jupyter for interactive access remote systems
jupyter
Web browser
End User
python

DEMO

Please, Click on Demo Video Link!

Q&A

THANK YOU!

Contact Seonggyu Park (seonggyu.park@brc.tamus.edu)

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- Jeffery Arnold (USDA)